

RESEARCH ARTICLE

Separations: Materials, Devices and Processes

In situ monitoring of CO₂ sorption on polyethylenimine dynamics through broadband dielectric spectroscopy

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Abstract

Chemical reactions between carbon dioxide (CO₂) and amine have been extensively characterized, however, their influence on the dynamics of polyamines remains largely unexplored. In this work, we compare the dynamics of polyethylenimine (PEI) before and after CO₂ absorption through broadband dielectric spectroscopy (BDS). The molecular processes of bulk PEI are very different from those of thin film PEI, highlighting an interesting interface and nano-confinement effect. Detailed analyses show CO₂ absorption slows down the PEI dynamics, which is consistent with an elevated glass transition temperature of PEI upon CO₂ absorption from differential scanning calorimetry measurements. Further *in situ* kinetic measurements demonstrate nonmonotonic changes in relaxation times or dielectric amplitudes of some relaxation processes during CO₂ sorption or desorption, suggesting an intriguing interplay between CO₂ chemisorption and the dynamics of PEI. These results demonstrate that BDS is a powerful platform to resolve the temporal dynamics changes of polyamines for CO₂ capture.

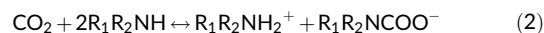
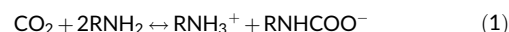
KEYWORDS

broadband dielectric spectroscopy, carbon capture, polyethyleneimine (PEI), polymer dynamics, thin polymer films

1 | INTRODUCTION

Carbon capture, especially carbon capture from dilute sources such as air, has the potential to address both current and legacy emissions to stabilize and even reduce the atmospheric carbon dioxide (CO₂) concentration.^{1–3} Sorbent-based technologies, such as polyamine films supported by mesoporous-substrates, have received significant attention due to their well-balanced CO₂-amine interaction suitable for CO₂ chemisorption at room or lower temperatures and CO₂ stripping at moderately elevated temperatures.^{4–10} However, existing polyamine systems suffer from several drawbacks due to the complex polyamine-air interactions and the air-induced irreversible or

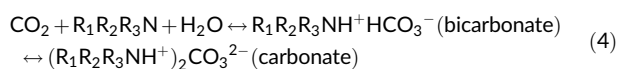
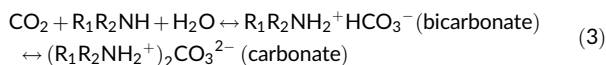
reversible chemical or physical changes. In particular, significant work has been conducted to study the CO₂-amine reactions.^{11–21} Under dry conditions, a reversible CO₂-amine reaction requires one CO₂ and two primary or secondary amines to give an ammonium carbamate:



When moisture (H₂O) is present, additional mechanisms are available as ammonium carbonate or ammonium bicarbonate form:

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We emphasize that the carbamic acid formation is the major initial PEI- CO_2 interaction process as the formation of bicarbonate is slow and requires high humidity.²² Although the influence of CO_2 on the chemical compositions of PEI has been extensively characterized, the impacts of CO_2 absorption on the dynamics of polyamines, such as poly(ethylenimine) (PEI) remain largely unexplored. Recent neutron scattering, nuclear magnetic resonance (NMR) measurements, and molecular dynamics simulations showed clear evidence of slowing down in the dynamics of small molecular amines or polyamines, for example the low molecular weight PEI, upon CO_2 sorption.^{23–26} The slowing down of PEI dynamics has also been found to have a profound influence on the kinetics of CO_2 sorption and desorption during carbon capture.²⁷ All these results suggest a strong impact on molecular interactions and the chemical compositions of PEI film upon CO_2 absorption as depicted in Figure 1. This highlights a crucial task of elucidating the dynamics of the sorbent, such as polyamines, with or without CO_2 sorption. However, a systematic study is missing that compares the dynamics of small molecular amines or polyamines in absence of CO_2 sorption to that with CO_2 sorption. Even less is known about the progressive changes to PEI dynamics during the CO_2 sorption or desorption. Part of this is because of the challenge of characterizing the time-resolved changes to the sorbent during CO_2 sorption or desorption, which requires the development of novel *in situ* experimental approaches.

Broadband dielectric spectroscopy (BDS) has been demonstrated as a robust method to characterize dynamics of polymers,²⁸ including polyamines.^{29,30} In a very recent study, we

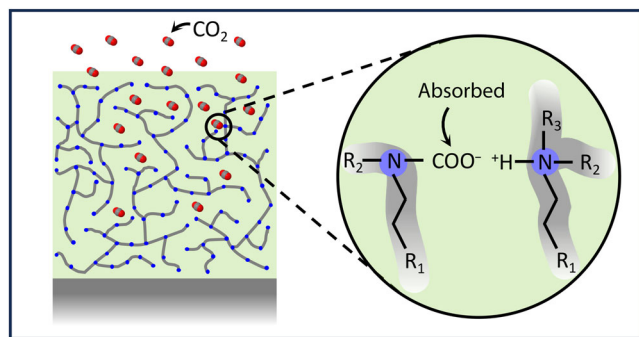


FIGURE 1 Schematic representations of a thin polyethylenimine (PEI) film on a substrate where solid grey lines are the PEI chains and the blue dots are the amine groups along the chains at which the absorption reaction takes place through ammonium carbamate formation following Equations (1) and (2) as depicted in the close-up. R_1 represents the continuing PEI chain while R_2 and R_3 are either PEI chains or terminal hydrogen depending on whether primary, secondary, or tertiary amines are involved.

have performed BDS measurements of one type of PEIs, PEI with a characteristic molecular weight of 800 g/mol, supported by a porous polyolefin substrate with a film of thickness of 10–100 μm .³¹ In particular, a large influence of CO_2 sorption on the glassy dynamics of these PEI films has been observed, implying a high sensitivity of BDS to quantify the influence of CO_2 sorption on the dynamics of PEI. However, in industrially relevant mesoporous-silica or other substrates supported polyamine systems, the thicknesses of the polyamine films are typically on the order of a few nanometers.^{1,10} On the other hand, more than three decades of intensive studies have found that the dynamics of thin polymer films down to less than 10–100 nm can be fundamentally different from that of the polymer film with thicknesses of 10 μm or more due to the interface effect and the nanoconfinement effect.^{32–38} Given the strong correlation between the dynamics and gas transport properties of polymers,^{27,39,40} understanding the impact of CO_2 on the molecular relaxations is critical for advanced carbon capture systems design. Therefore, it is desired to characterize the dynamics of PEI thin films on the order of several nanometers or tens of nanometers and study their interplay with CO_2 sorption or desorption. However, little has been unraveled about how the combined influences of the interface and the chemical reactions upon CO_2 sorption change thin PEI film dynamics.

Taking advantage of nanostructured electrodes, nano-BDS has been extended to quantify the dynamics of thin polymer films of less than 10 nm.^{41,42} Studies have also demonstrated that this nano-BDS platform can be modified to investigate the influence of solvent vapor on the dynamics of thin polymer films,⁴³ implying the potential of nano-BDS to serve as a platform to monitor *in situ* the dynamics of PEI thin films during CO_2 sorption and desorption. However, no work has been performed so far to characterize the dynamics of thin PEI films and to investigate the dynamics changes of PEI films during CO_2 sorption and desorption.

In this article, we aim to investigate the impact of CO_2 sorption or desorption on the dynamics of a model PEI with a molecular weight of 25 kg/mol (PEI25k) in both the bulk and thin film geometries through a combination of BDS and differential scanning calorimetry (DSC). We compare the dielectric spectra of the bulk and the thin film PEI around 25 nm in thickness under conditions of (i) saturated CO_2 sorption achieved through a long-time exposure of PEI to pure CO_2 atmosphere (CO_2 -PEI), (ii) a CO_2 desorbed, neat state created either by prolonged exposure to N_2 atmosphere at elevated temperature (N_2 -PEI) or in vacuum (v-PEI); further, (iii) a low to moderate CO_2 sorption was explored in some cases by exposition to the low CO_2 concentration of ambient air (air-PEI). Both the bulk and thin film CO_2 -PEI show significant slowing down in molecular dynamics compared with the respective v-PEI and N_2 -PEI. In addition, CO_2 -PEI also exhibits dramatically different dielectric dispersions and dielectric relaxation amplitudes from that of the v-PEI or N_2 -PEI. DSC measurements of bulk CO_2 -PEI show a noticeable increment in the glass transition temperature, T_g , compared with bulk v-PEI. All these observations highlight the large influence of CO_2 sorption on the dynamics of bulk PEI.

Furthermore, we perform *in situ* nano-BDS measurements on thin film PEI during CO₂ sorption and desorption for a time-resolved monitoring of the changes of their dielectric properties. Systematic and progressive changes of the dielectric dispersions of the thin film PEI have been quantified, which shows significant changes to the dynamics of the PEI film. In principle, these temporal dynamics changes correlate directly with the amount of CO₂ sorption or desorption and thus can be utilized to quantify the amount of CO₂ sorption or desorption during carbon capture. However, the main goal of this article is to reveal the large dynamics changes of bulk PEI and thin film PEI upon CO₂ sorption or desorption, and to demonstrate BDS as an effective analytical method to characterize them. We defer the investigations that map the dynamics changes with the amount of CO₂ sorption or desorption to a future study.

We note that in real carbon capture applications, PEI with lower molecular weights such as 800 g/mol (PEI800) or 1000 g/mol (PEI1000) were applied. In addition, the PEI films in these nanopores are typically around 10 nm thick or less, which is smaller than the 25-nm-film studied here. Since the fundamental PEI-CO₂ interactions do not change with the PEI molecular weights, we anticipate the obtained results from PEI25k to be relevant also for PEI800 or PEI1000. Our recent BDS studies³¹ on bulk PEI800 and the PEI800 thin film Figure S1 show qualitatively similar results as PEI25K, supporting the negligible effect of polymer molecular weights. In addition, a very recent study⁴⁴ concludes that the CO₂ capture is limited by the CO₂ availability as long as the film thickness is less than 50 nm. The results thus suggest that studies of 25 nm film can be informative for thinner films, while more work needs to be performed on this topic. Therefore, we believe the presented results demonstrate unambiguously that BDS can serve as a robust platform to monitor the CO₂ capture performance of polyamine-based systems.

2 | MATERIALS AND METHODS

2.1 | Materials

Branched PEI with a weight-averaged molecular weight of 25 kg/mol and polydispersity index (PDI) of 2.5 was purchased from Sigma-Aldrich (Product Number: 408727) and used after drying for 24 h in a vacuum oven at $T = 343$ K and a vacuum level of 10^{-3} torr. The bulk v-PEI was prepared at $T = 343$ K under vacuum for 24 h. The bulk CO₂-PEI was prepared by exposing the bulk v-PEI under a pure CO₂ atmosphere (partial pressure ~ 1 atm) at $T = 343$ K for at least 12 h. We have measured the BDS and the DSC of both the bulk v-PEI and bulk CO₂-PEI. To prepare thin film N₂-PEI, 20 mg PEI was dissolved in 20 mL ethanol and spin-cast onto silicon substrates at 1200 rpm, which gave a thin film of a thickness of around 25 nm as confirmed by our atomic force microscopy (AFM) measurements. Prior to measurements, the thin film PEI was annealed at $T = 340$ K in pure N₂ atmosphere in the dielectric spectrometer and monitored until a stable signal was reached. As demonstrated in the following sections, the segmental relaxation time, τ , of thin film CO₂-PEI is around 10^{-4} –

10^{-5} s at $T = 340$ K. This leads to a characteristic mean-free diffusion length of 6 – 10 μm for 1 h of annealing at $T = 340$ K, which is much larger than the film thickness of 25 nm. Thus, CO₂ or N₂ can readily access the whole film before the tests.

2.2 | Differential scanning calorimetry

DSC has been employed to quantify the glass transition temperature of bulk v-PEI and bulk CO₂-PEI. For bulk v-PEI, 10 mg of bulk PEI was filled in an Aluminum pan that was degassed in a vacuum oven (10^{-3} torr) at room temperature for 24 h before being capped with an aluminum lid. To obtain bulk CO₂-PEI, the uncapped v-PEI pan was placed in a sealed aluminum can filled with pure CO₂ with a pressure of 1.07 bar at $T = 343$ K for 12 h before being capped with a lid. A heating and cooling rate of 5 K/min was applied for the DSC measurements. The glass transition temperature was identified through the inflection point of the specific heat flow curve.

2.3 | Dielectric spectroscopy

Dielectric spectroscopy was applied to quantify the dynamics of bulk PEI and thin film PEI on a Novocontrol Concept 40 with an Alpha Analyzer and a Quatro Cryosystem temperature control system. The temperature accuracy is 0.1 K. For the dynamics of bulk PEI, a thin layer of bulk PEI was deposited onto the surface of a 20-mm-diameter gold electrode and confined by a hollow Teflon ring spacer with a thickness of 0.14 mm, an inner diameter of 14 mm, and an outer diameter of 26 mm. The bulk v-PEI was then dried in a vacuum oven (10^{-3} torr) for 24 h before being capped with a gold top electrode of 20-mm diameter. The bulk CO₂-PEI was prepared by exposing the uncapped bulk v-PEI in a sealed aluminum can to pure CO₂ of a pressure of around 1.07 bar for 12 h. Then, the CO₂ saturated PEI was capped with a top gold electrode with a hollow Teflon ring spacer that seals the side of the electrode/PEI/electrode sandwich structures and prevents the escape of CO₂ during dielectric measurements. The bulk PEI has a segmental relaxation time of around 10^{-7} s at $T = 293$ K. The 12-h exposure to CO₂ at $T = 293$ K thus gives a CO₂ mean-free path of approximately 0.5–1.0 mm, which is much larger than the film thickness of 0.14 mm. Therefore, the 12-h annealing is sufficient to reach an equilibrium state of CO₂ sorption across the film. The measurements were performed at a frequency range between 10^7 and 10^{-2} Hz and a temperature range between 200 and 340 K at an interval of 10 K upon heating. A thermal equilibration of 20 min was applied at each temperature before measurements. All the measurements were performed in a nitrogen atmosphere unless specified otherwise and the average mean-squared voltage was set to be 0.1 V.

For dielectric measurements of thin films, ultra-flat (RMS roughness ≤ 0.5 nm) and highly conductive (specific resistance ≤ 3 m Ωcm) silicon electrodes served as substrates (Figure 2A). The spin-cast PEI layer of about 25-nm thickness (determined by AFM after the

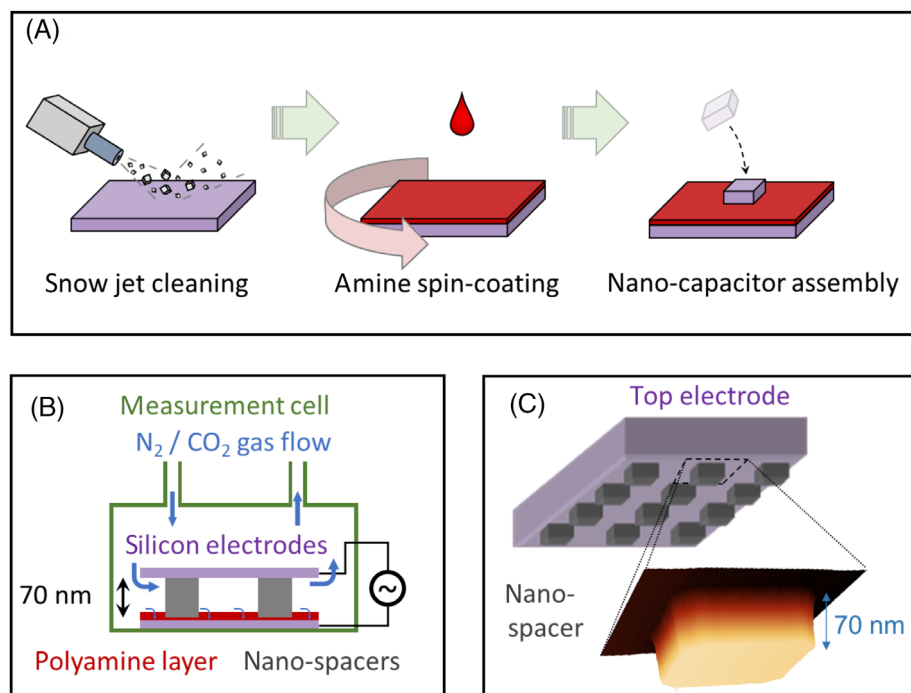


FIGURE 2 (A) A sketch of the preparation steps for broadband dielectric spectroscopy (BDS) measurements of the thin film (see text for details). (B) A schematic representation of the thin film sample in the BDS measurement cell allowing a variation of the gas exposure. (C) A sketch of the top electrode with its regular array of nano-spacers and an atomic force microscopy (AFM) image showing a single spacer. Note that the nano-spacers are separated stilts as shown in panel (C); thus, they do not block the gas flow and the CO₂ can access whole polyethylenimine film.

dielectric measurements) was then covered by a top electrode made of doped silicon which is covered with a matrix of insulating silica nano-structures (Figure 2B,C) acting as spacers. These spacers have a height of 70 nm, a base area of $5 \times 5 \mu\text{m}$, and are separated by $45 \mu\text{m}$ on a square grid (hence, the area and volume of the sample capacitor occupied by silica is $\approx 1\%$). It is worth noting that these nanospacers are separated stilts and do not block the pathway of lateral gas flow through the surface of the PEI film as well as the free volume above the film. This capacitor with an electrode separation on the nanometer scale was then placed in a sample cell (Figure 2B) which had been developed to study *in situ* solvent vapor exposition of polymer brushes during dielectric measurements.⁴³ This setup was modified to expose the thin PEI film to ambient atmosphere, pure nitrogen, or pure CO₂, respectively, while also controlling the temperature in the BDS spectrometer. These measurements were conducted with a similar spectrometer to the one used for the bulk measurements in a frequency range of 10^6 to 10^{-1} Hz and at temperatures between 200 and 360 K.

To avoid excessive and potentially destructive electrical fields (due to the small electrode spacing), the AC voltage amplitude was limited to 0.1 V in these measurements. For thin films, two types of experiments were performed: (i) For dynamics examination similar to the bulk measurements, isothermal spectra were collected at a range of different temperatures (in increments of 2 K or 5 K) of thin PEI films that were brought into a completely adsorbed or desorbed state by exposing them for longer periods to either CO₂ (thin film CO₂-PEI) or N₂ (thin film N₂-PEI), respectively; to assist the desorption and to avoid the influence of PEI crystallization, the temperature was risen to 340 K. (ii) Kinetic studies of the adsorption and desorption processes were performed through continuous recording of spectra at T

$= 340$ K over time during and after switching from one type of gas to another one.

3 | RESULTS AND DISCUSSIONS

3.1 | Bulk PEI

Figure 3 depicts the dielectric derivative spectra $\epsilon'_{\text{der}}(\omega)$ (main frame) and the corresponding storage permittivity (filled symbols), $\epsilon'(\omega)$, and loss permittivity (open symbols) $\epsilon''(\omega)$ (inset) of the bulk v-PEI (Figure 3A) and the bulk CO₂-PEI (Figure 3B) at several representative temperatures. The former was employed in the past to improve the identification of relaxations because of its sharper peaks as well as the absence of DC-conductivity contribution at low angular frequencies ω ⁴⁵

$$\epsilon'_{\text{der}}(\omega) = -\frac{\pi}{2} \frac{\partial}{\partial \ln \omega} \epsilon'(\omega), \quad (5)$$

where $\ln$ is the natural logarithm and ∂ is the partial differential.

Due to the strong DC-conductivity contributions, the dielectric relaxation processes of both the bulk v-PEI and bulk CO₂-PEI are more readily seen in the derivative spectra (Figure 3). In both cases, a broad dielectric peak has been observed at higher frequencies followed by a sharp and well-resolved relaxation peak at low frequencies. Interestingly, the dielectric dispersion of bulk CO₂-PEI is significantly broader than that of the bulk v-PEI. The shapes of the dielectric dispersion contain information on the dynamics heterogeneity. The broader the dielectric dispersion, the more

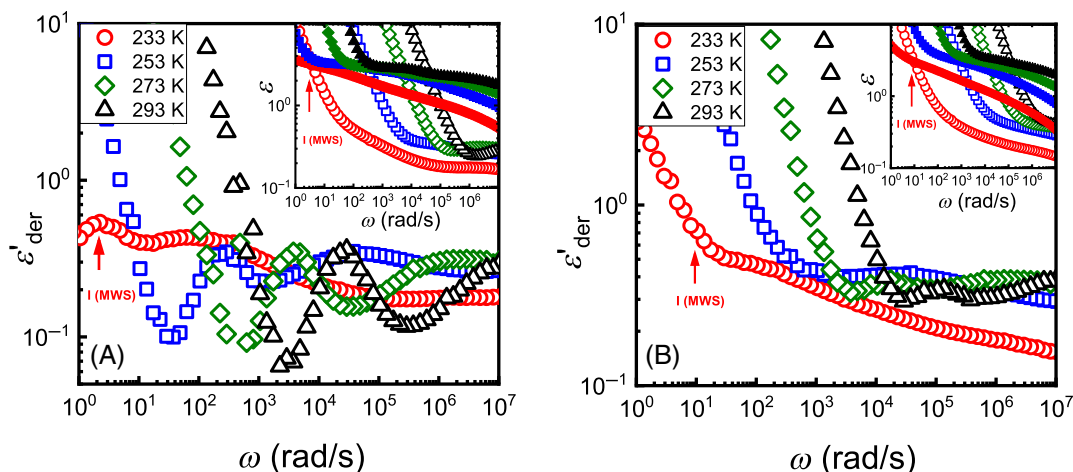


FIGURE 3 Derivative spectra $\epsilon'_{\text{der}}(\omega)$ of (A) bulk v-polyethylenimine (PEI) and (B) bulk CO_2 -PEI at different temperatures as indicated. The insets depict the corresponding dielectric spectra $\epsilon''(\omega)$ (open symbols) and $\epsilon'(\omega)$ (filled symbols). The experimental uncertainty is smaller than the symbol size.

heterogeneous the dynamics. These observations thus imply the CO_2 absorption generates a more heterogeneous local dynamics environment in bulk PEI, which is anticipated since the absorption of CO_2 creates additional interatomic interactions between amine groups and the CO_2 as well as long-range Coulombic interactions (Figure 1) that do not exist in v-PEI.

The low-frequency sharp peaks are located at around the characteristic frequency ω_c of $\epsilon'(\omega_c) \approx \epsilon''(\omega_c)$ (see the red arrows of Figure 3 at $T = 233$ K), indicating its Maxwell-Wagner-Sillars (MWS) interfacial polarization in origin.^{34,46,47} Note that the MWS processes of bulk CO_2 -PEI at low temperatures, for example at $T = 253$ K or $T = 233$ K, are superimposed by the electrode polarization in the derivative spectra and the DC-conductivity in the loss spectra (inset of Figure 3B). The MWS processes are typically due to interfacial polarization that can happen at the interface between the polymer matrix and the impurities and is not associated with any molecular relaxation process.²⁸ Thus, we do not analyze them in this work since the primary focus of this work is molecular dynamics. Furthermore, the dielectric spectra of bulk CO_2 -PEI show excellent reversibility upon heating and cooling at all frequencies (see Figure S2), indicating negligible CO_2 desorption during the dielectric measurements.

To analyze quantitatively the influence of CO_2 adsorption on the dynamics of bulk PEI, the complex dielectric permittivity spectra $\epsilon^*(\omega) = \epsilon'(\omega) - i\epsilon''(\omega)$ with $\epsilon'(\omega)$ and $\epsilon''(\omega)$ being the storage and the loss permittivity, and the imaginary unit $i = \sqrt{-1}$, are fitted with a sum of several Havriliak Negami (HN) functions,⁴⁸ and, if necessary, a power law describing a conductivity contribution in the imaginary part⁴⁹ and the electrode polarization in the real part.⁵⁰

$$\epsilon'(\omega) = \epsilon_\infty + \sum_{j=1}^n \text{Re} \left(\frac{\Delta\epsilon_j}{(1 + (i\omega\tau_{\text{HN}j})^{\beta_j})^{\gamma_j}} \right), \quad (6)$$

$$\epsilon''(\omega) = \sum_{j=1}^n \text{Im} \left(\frac{\Delta\epsilon_j}{(1 + (i\omega\tau_{\text{HN}j})^{\beta_j})^{\gamma_j}} \right) + \frac{\sigma_{\text{DC}}}{\epsilon_0\omega}, \quad (7)$$

where $\Delta\epsilon_j$, $\tau_{\text{HN}j}$, β_j , and γ_j are the dielectric relaxation strength, the characteristic HN relaxation time, and the symmetric and asymmetric broadening parameters of the j th relaxation process, respectively; ϵ_0 and σ_{DC} are the permittivity of vacuum, and the DC-conductivity, respectively. For a most exhaustive use of the data, three data sets— $\epsilon'(\omega)$, $\epsilon''(\omega)$ and ϵ'_{der} —were fitted with the same parameter set. For the latter, a modified HN function was used⁴⁵

$$\epsilon'_{\text{der}}(\omega) = - \frac{\beta\gamma\Delta\epsilon(\omega\tau_{\text{HN}})^{\beta} \cos\left(\frac{\pi\beta}{2} - (1+\gamma)\arctan\left[\frac{\sin(\frac{\pi\beta}{2})}{(\omega\tau_{\text{HN}})^{-\beta} + \cos(\frac{\pi\beta}{2})}\right]\right)}{\left[1 + 2(\omega\tau_{\text{HN}})^{\beta} \cos\left(\frac{\pi\beta}{2}\right) + (\omega\tau_{\text{HN}})^{2\beta}\right]^{\frac{1+\gamma}{2}}} \quad (8)$$

Each HN function corresponds to a certain polarization relaxation process. The characteristic relaxation time of each polarization relaxation process can be obtained through

$$\tau_c = \tau_{\text{HN}} \left[\sin\left(\frac{\pi\beta}{2+2\gamma}\right) \right]^{-\frac{1}{\beta}} \left[\sin\left(\frac{\pi\beta\gamma}{2+2\gamma}\right) \right]^{\frac{1}{\beta}}. \quad (9)$$

Figure 4A,B provides a representative fit to the derivative (main frame) and loss (inset) spectrum of bulk v-PEI and bulk CO_2 -PEI at $T = 253$ K. Due to the broad dielectric dispersions, three HN functions (the dashed-dot lines, the dashed lines, and the short dotted lines in Figure 4A,B) are required to fit the spectra, including the MWS process at the low frequencies. The fit has further confirmed the much broader dielectric dispersions of bulk CO_2 -PEI than the bulk v-PEI. To facilitate the discussion, we name the dielectric relaxation processes from low frequency to high frequency as process I, II, and III. In particular, process I is the MWS contribution due to its sharp dielectric dispersion and the characteristic frequency coinciding with the location

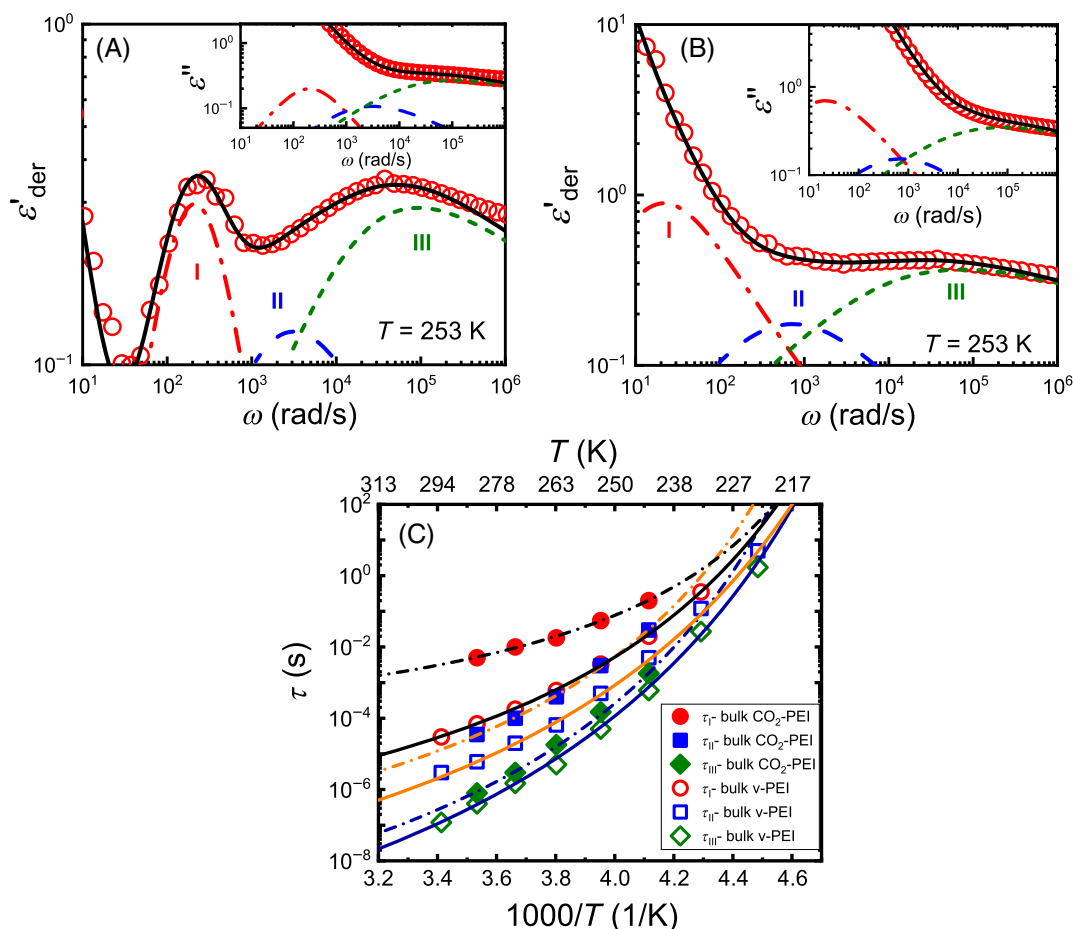


FIGURE 4 Derivative spectra $\epsilon'_{\text{der}}(\omega)$ of (A) bulk v-polyethylenimine (PEI) and (B) bulk CO₂-PEI at a temperature of 253 K. The insets depict the corresponding dielectric loss spectra $\epsilon''(\omega)$. The black solid line is a fit to Equation (8) (Equation 7 for the insets), and the red dash-dotted, blue dashed, and green dotted lines are the individual Havriliak Negami (HN) functions. Panel (C) depicts the characteristic relaxation time τ (calculated from the HN fit parameters via Equation 9) for processes I, II, and III in bulk v-PEI (open symbols) and CO₂-PEI (filled symbols); the solid and dash-dotted lines are VFT-fits (Equation 10) to v-PEI and CO₂-PEI, respectively, where black, orange, and dark blue refer to processes I, II, and III. The experimental uncertainty is smaller than the symbol size.

at $\epsilon'(\omega) \approx \epsilon''(\omega)$. With the complex molecular structure of branched PEI and the rich local interactions between primary, secondary, and tertiary amine groups as well as their active interaction with CO₂, additional measurements, such as NMR characterization, may help to reveal the explicit molecular origins of the processes II and III.

To further understand the molecular origin of these relaxations, we analyze the temperature dependence of its relaxation time. Figure 4C presents the temperature dependence of relaxation times, where the open symbols are the relaxation times of bulk v-PEI and the filled symbols are those of the bulk CO₂-PEI. The relaxation times of bulk CO₂-PEI are much slower than those of bulk v-PEI, highlighting the impact of CO₂ adsorption on PEI dynamics. Furthermore, processes I, II, and III exhibit different degrees of slowing down with the same CO₂ adsorption, clearly encoding peculiarities of their distinct molecular mechanism. Generally, localized and cooperative relaxations are distinguished where the former follows an Arrhenius law while the latter does not; the super-Arrhenius dependence is usually described by the empirical Vogel-Fulcher-Tammann (VFT) equation^{51–53}

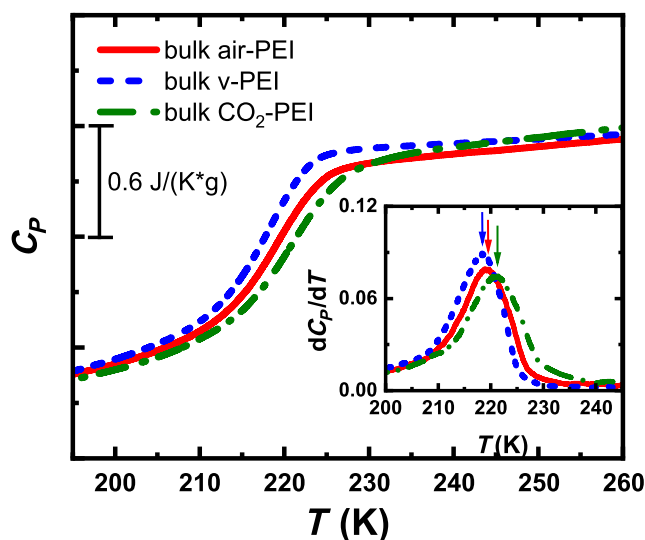
$$\tau(T) = \tau_0 \exp\left(\frac{DT_0}{T - T_0}\right), \quad (10)$$

where τ_0 is the characteristic relaxation time in the high-temperature limit, D is a constant related to the fragility, and T_0 is the so-called Vogel temperature. The molecular origin of the super-Arrhenius temperature dependence of relaxation times has been a topic of active discussion.^{54–56} However, a general view is that processes which require molecular cooperativity follow a super-Arrhenius temperature dependence as the number of molecules involved cooperatively increases with cooling.^{57,58} All observed processes I, II, and III exhibit a VFT dependence rather than an Arrhenius behavior (Figure 4C) and we present the fit parameters in Table 1. These detailed analyses suggest that none of these processes originate from localized fluctuations. Instead, cooperative processes, such as structural relaxation also called α -relaxation, are candidates characterized by a VFT dependence. In particular, the structural relaxation by convention coincides with the glass transition temperature at a timescale of ≈ 100 s.^{59,60}

TABLE 1 VFT fit parameters (Equation 10) to the different relaxations identified in the dielectric measurements as well as the estimated T_{100} of process III.

Sample	$\tau_{0,I}$ (s)	D_I	$T_{0,I}$ (K)	$\tau_{0,II}$ (s)	D_{II}	$T_{0,II}$ (K)	$\tau_{0,III}$ (s)	D_{III}	$T_{0,III}$ (K)	T_{100} (K)
bulk v-PEI	5.6×10^{-9}	5.6	177.6	3.9×10^{-11}	7.9	170	3.7×10^{-13}	9.2	170	217
bulk CO ₂ -PEI	3.4×10^{-5}	2.54	188.2	9.4×10^{-11}	8.3	171	5.1×10^{-12}	6.9	180	221
thin film N ₂ -PEI	2.3×10^{-4}	0.08	308	9.8×10^{-6}	0.38	285	1.2×10^{-8}	0.9	268	278
thin film CO ₂ -PEI	6.8×10^{-4}	0.006	318	8.4×10^{-5}	0.04	311	6.1×10^{-7}	0.1	311	312

Note: For the thin film measurements, the data points gathered during the cooling run were used.

**FIGURE 5** Specific heat capacity of bulk v-polyethylenimine (PEI), bulk CO₂-PEI, and bulk air-PEI as a function of temperature recorded at a rate of 5 K min⁻¹. The inset shows the derivatives of these curves with respect dC_p/dT to temperature; the vertical arrows indicate the respective peak position, which was taken as glass transition temperature T_g .

Consequently, employing Equation (10) for the extrapolation $\tau(T_{100}) = 100$ s yields an estimated T_{100} for the corresponding relaxation process. A comparison between T_{100} and the glass transition temperature from other measurements, such as DSC, can help identify the structural relaxation process.

To confirm the structural relaxation of bulk PEI and to characterize the influence of CO₂ absorption on the thermodynamic properties of bulk PEI, we have further performed DSC measurements where the glass transition temperature (T_g) can be identified independently from thermodynamics measurements. Figure 5 presents the specific heat capacity, C_p , and its derivative with respect to temperature (inset Figure 5), dC_p/dT . No crystallization, cold recrystallization, and melting have been observed, probably due to the branching structures, which is consistent with the previous literature reports.^{29–31} The glass transition step is visible with a characteristic glass transition temperature of $T_g = 218$ K and $T_g = 221$ K for bulk v-PEI and bulk CO₂-PEI, respectively (the arrows in the inset of Figure 5), which roughly correspond to the values of T_{100} of their corresponding process III. The

DSC T_g and BDS T_{100} of process III values of bulk PEI also agree well with previous literature reports of the T_g of PEI25k at about 221 K.⁶¹ Similar to the BDS results, the DSC data reveal an increment of T_g due to CO₂-sorption of bulk PEI. This confirms process III as the structural relaxation of bulk PEI. Furthermore, the C_p step of bulk CO₂-PEI is significantly broader than the one in bulk v-PEI, especially at the high-temperature side, implying a higher dynamical heterogeneity of bulk CO₂-PEI compared to bulk v-PEI. This is consistent with the broader dielectric dispersions of bulk CO₂-PEI (Figure 4). Regarding the molecular origin of processes II, we can only speculate at this moment. Possible assignments might be secondary or tertiary amine units since these can be expected to exhibit slower dynamics due to the increased number of connecting chains, or association/ dissociation dynamics of NH hydrogen bonds which are also dielectrically active.^{62,63} Nevertheless, these results demonstrate the sensitivity of BDS as an alternative platform to characterize the influence of CO₂ absorption on PEI dynamics. Furthermore, recent theories^{39,40,64} suggest the strong coupling between the structural relaxation of polymers and their gas transport properties. Thus, the lack of molecular assignment of processes I and II should not affect the follow-up discussions as our emphasis is on structural relaxation, for example, the process III.

3.2 | PEI thin film

3.2.1 | Dielectric properties of PEI thin films in absence of CO₂ adsorption

To obtain the dielectric properties of the thin film N₂-PEI, we have exposed the spin-cast PEI at $T = 340$ K for two hours to continuous N₂ flow. As demonstrated later in the kinetics section, the 2-h exposure in N₂ flow at $T = 340$ K is sufficient to eliminate the influence of air (primarily CO₂) on the dynamics. After the 2-h exposure to N₂ flow, we have performed BDS measurements of this supported PEI thin film under N₂ flow at different temperatures using setup described in Figure 2B from $T = 340$ K to $T = 300$ K at an interval of 5 K.

Figure 6A presents selected dielectric loss spectra $\epsilon''(\omega)$ of this PEI thin film where several interesting features are worth noting: (i) Multiple dielectric relaxation processes can be observed at all these high temperatures in the frequency window of 10^2 – 10^6 Hz,

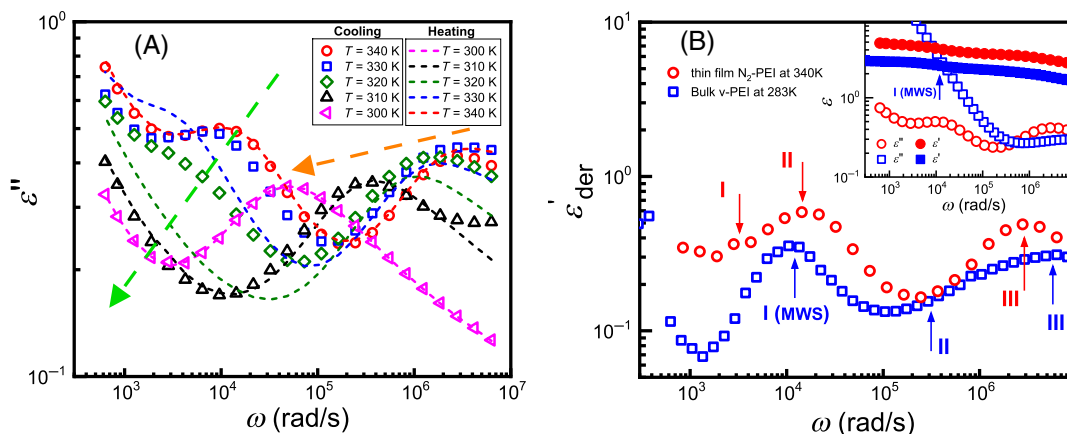


FIGURE 6 (A) Dielectric loss spectra $\varepsilon''(\omega)$ of thin film N_2 -polyethylenimine (PEI) at different temperatures as indicated upon cooling (open symbols) and heating (dashed lines). The green and orange arrows indicate the evolution of the mean relaxation rate and the dielectric relaxation strength of processes II and III, respectively, upon cooling. Panel (B) depicts a comparison of the derivative spectra $\varepsilon'_{\text{der}}(\omega)$ of the thin film N_2 -PEI at 340 K along with the bulk v-PEI at 283 K to demonstrate the coincidence of process III which highlights the large shift of thermal activation of the dynamics; the inset presents the corresponding loss spectra.

corresponding to molecular dynamics at $\approx 1.6 \times 10^{-3}$ s to $\approx 1.6 \times 10^{-7}$ s. For instance, the green and the orange arrows of Figure 6A give the changes of two individual relaxation processes with cooling. (ii) The dielectric relaxations of the PEI thin film exhibit distinct behavior in two temperature ranges, one above 325 K and one below. Specifically, the amplitudes of the dielectric dispersions at $T < 325$ K are significantly smaller than those at $T > 325$ K. The dielectric loss spectra upon heating (the dashed lines in Figure 6A) overlap excellently with the ones upon cooling (the symbols of Figure 6A) at the same temperatures well below 325 K (such as at $T = 300$ K) and well above 325 K (such as at $T = 340$ K), suggesting reversible changes of the thin film N_2 -PEI upon heating and cooling. (iii) Furthermore, at $T = 320$ K and $T = 330$ K the dielectric spectra upon heating (dashed lines) do not coincide with the ones upon cooling (symbols), indicating the presence of a hysteresis of dielectric properties at temperatures around 325 K. One can further observe that at around $T = 325$ K the dielectric relaxation strengths of all relaxation processes upon heating are much smaller than those during cooling. These features all point to a reversible transition in the PEI film.

One possibility might be residual moisture due to the hydrophilic nature of PEI. However, our infrared (IR) spectroscopy measurements in both the near IR and the mid-IR region show little signs of water (see Figure S3). We further note that a recent study⁴⁴ prepared thin PEI25k films the same way as presented here. The authors found no residual moisture or solvent during the *in situ* IR measurements, which agrees well with our observations. Additionally, the nice overlap of the dielectric spectra upon subsequent heating and cooling also indicates the absence of any dynamical changes due to potential residual moisture and solvent. We note further that short-chain linear PEI has a melting temperature between 284 and 331 K,^{30,65,66} which might be the origin of the changes in dielectric spectra at around $T = 325$ K. However, this assignment is ambiguous since we have not seen any sign of crystallization in our DSC measurements for bulk PEI (Figure 5). These observations might indicate a weak tendency to

crystallize in the branched PEI or some intriguing role of the interface in the crystallization of PEI thin films. Given the uncertainty of the dynamics changes at $T = 325$ K and the robustness of the dielectric properties of PEI film above $T = 325$ K, we examine the capability of the BDS platform for CO_2 sorption and desorption at a higher temperature, specifically at $T = 340$ K.

In addition, we compare the dielectric spectra of the thin film N_2 -PEI at $T = 340$ K with the bulk v-PEI at $T = 283$ K as shown in Figure 6B, where the derivative spectra are provided in the mainframe and the storage (filled symbols) and the loss (open symbols) permittivities are in the inset. The characteristic peak of the primary relaxation process of bulk v-PEI at $T = 283$ K (Process III) is comparable with the main relaxation process of the thin film N_2 -PEI at $T = 340$ K, suggesting a strong increment in the glass transition temperature of the thin film N_2 -PEI. The observed large shift in dynamics in thin film N_2 -PEI is expected due to the strong Hydrogen bonding interactions between PEI and the surface of the electrodes. This observation is also important to understand the carbon capture performance since the sorbent in almost all mesoporous silica-supported PEI systems is in a thin-film-form.

Process II of the thin film N_2 -PEI is well-separated from that of the primary relaxation, while process II and process III of the bulk v-PEI are strongly overlapping. Note that the sharp relaxation process of bulk v-PEI at low frequencies is the MWS process of the bulk v-PEI, which takes place at the crossover between the storage and loss permittivity (see the arrow in the inset of the Figure 6B). In contrast, the crossover frequency of the storage and loss permittivity of the thin film N_2 -PEI appears at much lower frequencies outside the frequency window of the measurements (the inset of the Figure 6B). The differences in dielectric shapes between the thin film N_2 -PEI and the bulk v-PEI thus point out another major feature of their dynamics: the large separation in relaxation processes of thin film N_2 -PEI indicates an intriguing interface effect that affects the various relaxation processes of PEI differently. These measurements also imply a

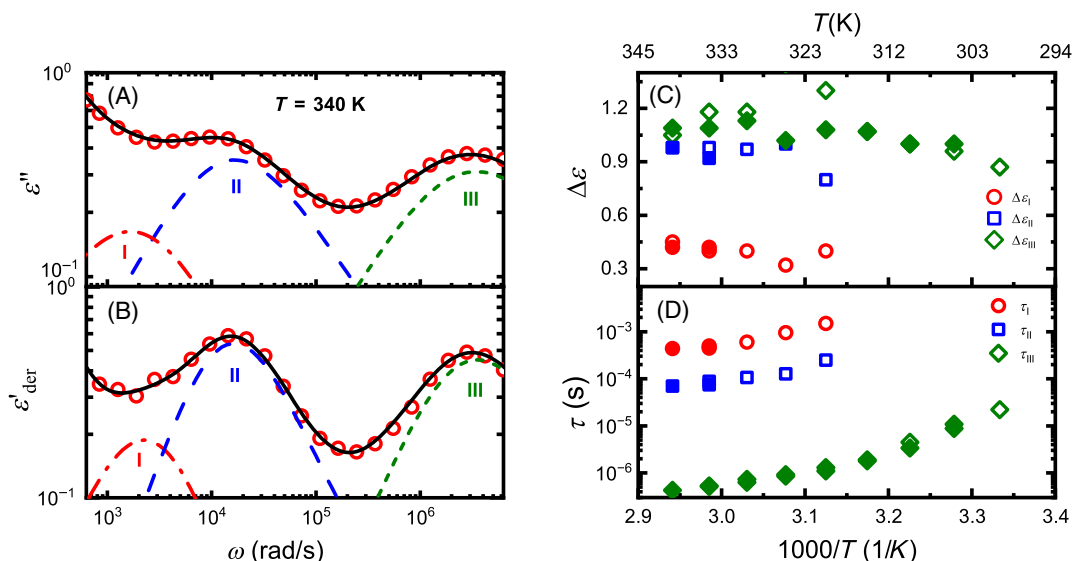


FIGURE 7 (A) Dielectric loss spectra $\varepsilon''(\omega)$ and (B) derivative spectra $\varepsilon'_{\text{der}}(\omega)$ thin film N_2 -polyethylenimine (PEI) at 340 K. The black solid line is a fit to Equations (7) (Panel a) and 8 (Panel b), and the red dash-dotted, blue dashed, and green dotted lines are the individual HN functions of process I, II, and III, respectively. (C) Dielectric relaxation strength $\Delta\varepsilon(T)$ and (D) characteristic relaxation time $\tau(T)$ of thin film N_2 -PEI for process I, II, and III, respectively, upon cooling (open symbols) and heating (filled symbols).

potentially large effect of the film thickness on polymer dynamics that is known to impact strongly the gas transport.^{39,64}

To understand the dielectric relaxations of the thin film N_2 -PEI, we fit the dielectric spectra with a sum of HN functions (Equation 7). Figure 7 provides a representative dielectric spectrum of thin film N_2 -PEI along with its fit. At least three HN functions are needed to describe fully the dielectric spectra. To address these three relaxations, we assigned processes I, II, and III from low frequencies to high frequencies. From the fit, one obtains the dielectric relaxation strengths (Figure 7C), $\Delta\varepsilon_{\text{I}}$, $\Delta\varepsilon_{\text{II}}$, and $\Delta\varepsilon_{\text{III}}$ as well as the corresponding characteristic relaxation times (Figure 7D), τ_{I} , τ_{II} , and τ_{III} .

For processes I and II, within the limited temperature and frequency ranges, one does not see much difference in their relaxation strengths (Figure 7C). However, a clear increment upon cooling was observed for process III at $T > 325$ K, which reduces upon further cooling at $T < 325$ K. At the same time, a remarkable hysteresis has been observed at $T \sim 325$ K in $\Delta\varepsilon_{\text{III}}$ between cooling and heating. These observations align with our initial conjecture of certain phase transition that alters the dielectric properties. On the other hand, the relaxation times of process III, τ_{III} , increase strongly upon cooling with a pronounced super-Arrhenius manner, suggesting a collective nature of process III. This process is most likely the primary (i.e., structural) relaxation of the thin film N_2 -PEI associated with the glass transition. Unfortunately, we only get limited results on τ_{I} and τ_{II} at the frequency and temperature window of the measurements. Hence, their molecular origin remains unresolved. Since processes I and II are slower than the main relaxation process III, it is unlikely that they belong to some secondary (i.e. localized) relaxation. Given their dielectric activities, we may speculate that they correlate with molecular dynamics involving amine groups.⁶⁷ However, further investigations

are required for a more detailed assignment of these dielectric relaxation modes.

3.2.2 | The influence of CO_2 sorption on the dielectric properties of PEI thin films

To characterize the influence of CO_2 sorption on the dynamics of the thin PEI film, we have further performed BDS measurements and analyzed the dielectric spectra of thin film CO_2 -PEI, that is, measurements were performed after exposure of the PEI film to the CO_2 atmosphere for at least 2 h and the measurements were performed during an ongoing CO_2 stream. Figure 8 shows representative dielectric spectra of thin film CO_2 -PEI upon cooling (symbols), and heating (dashed lines) in the range from $T = 300$ K to $T = 340$ K. One important observation is the good agreement of the dielectric spectra of thin film CO_2 -PEI at $T = 300$ K and $T = 340$ K upon cooling and heating, suggesting the reversibility of the dielectric measurements and little permanent changes of the PEI film at the testing temperatures. Interestingly, we have also observed a sudden change in the relaxation shapes and amplitudes at around $T \approx 320$ K, which divides the dielectric properties of thin film CO_2 into two distinct groups. Additionally, a hysteresis in dielectric properties of PEI film with absorbed CO_2 has also been observed at around $T = 320$ K. All these observations are consistent with that of the N_2 -PEI film and point to the existence of certain phase transitions at $T \approx 320$ K for thin film CO_2 -PEI. Interestingly, the temperature range of the hysteresis of thin film CO_2 -PEI is around $T \approx 320$ K which is slightly lower than the hysteresis temperature of $T \approx 325$ K for thin film N_2 -PEI. These observations thus highlight the influence of CO_2 absorption on the reversible transition of thin PEI film, whose origin remains further investigation.

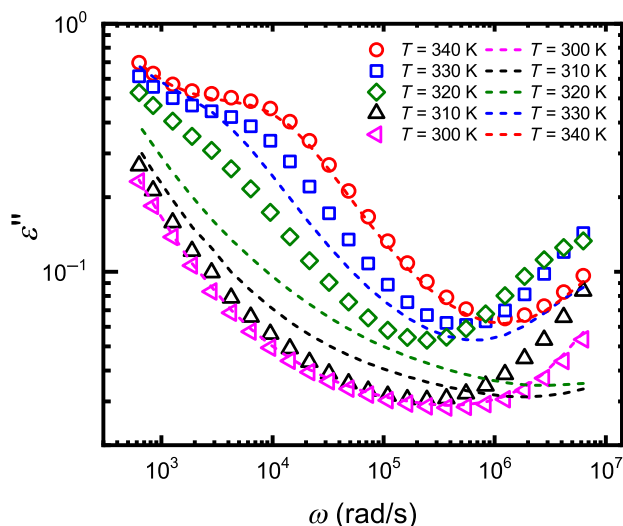


FIGURE 8 Dielectric loss spectra $\varepsilon''(\omega)$ of thin film CO_2 -PEI at different temperatures as indicated upon cooling (open symbols) and heating (dashed lines).

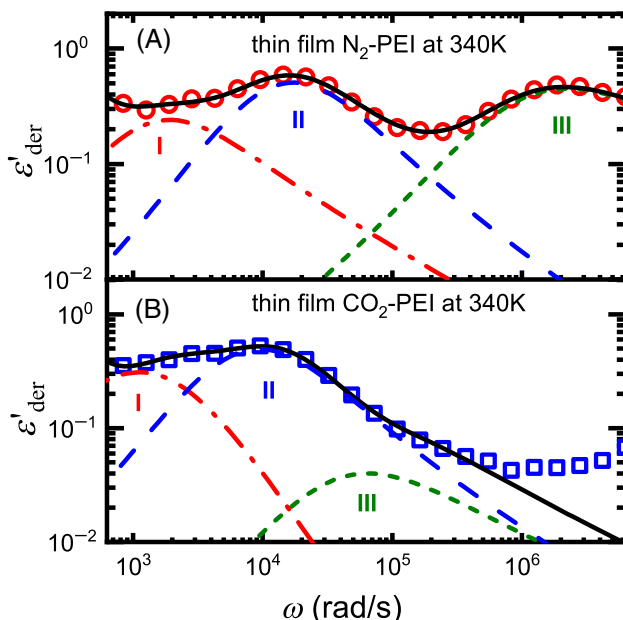


FIGURE 9 Derivative spectra $\varepsilon'_{\text{der}}(\omega)$ of (A) thin film N_2 -polyethylenimine (PEI) and (B) thin film CO_2 -PEI at 340 K. The black solid lines are fits to Equation (8), and the red dash-dotted, blue dashed, and green short dashed lines are the individual Havriliak-Negami (HN) functions of process I, II, and III, respectively.

Figure 9A gives a representative HN function fit to the dielectric spectra of the thin film CO_2 -PEI at $T = 340$ K. We have also included the dielectric properties of thin film N_2 -PEI at the same temperature for comparison (Figure 9B). The sum of these HN functions is also given (black solid line), while the dash-dotted, dashed, and short dashed lines represent processes I, II, and III, respectively, from low frequencies to high frequencies. Note that in the thin film setup, the slightly limited conductivity of the silicon electrodes (compared to

metal) causes a high-frequency contribution which was approximated with an additional relaxation process for a formal signal description (see the magenta dotted line in Figure 9B),⁴³ that is, it is not considered for further analysis. Compared with thin film N_2 -PEI, the thin film CO_2 -PEI exhibits a slowing down in the primary relaxation (Process III). In particular, the characteristic relaxation time of process III at $T = 340$ K shifts from 3×10^{-7} s in thin film N_2 -PEI to 2×10^{-5} s in thin film CO_2 -PEI, highlighting the influence of CO_2 absorption on the dynamics of PEI. In addition, with CO_2 sorption the dielectric relaxation strengths of the primary relaxation of PEI films are also reduced, which is consistent with the restricted motion of the amine groups due to CO_2 sorption. These observations are thus in agreement with the strong interaction between CO_2 and the PEI and further confirm a high sensitivity of BDS to reveal the influence of CO_2 absorption on dynamics of thin PEI films.

3.2.3 | The kinetics of CO_2 desorption and absorption

The large influence of CO_2 sorption on the thin PEI film dynamics as well as the high sensitivity of BDS to characterize their dynamics inspires further exploration of the potential of BDS to monitor in situ the kinetics of CO_2 desorption and absorption. To obtain the kinetics of CO_2 desorption, one can characterize the time-resolved dielectric properties of thin film CO_2 -PEI under a continuous N_2 environment.

Since the changes in dielectric spectra of PEI films are due to CO_2 desorption or absorption, quantitative descriptions of the dielectric spectra change thus infers the CO_2 desorption and absorption. To achieve that, a one-on-one mapping between specific CO_2 loadings with the relaxation time or dielectric relaxation amplitude and the CO_2 sorption-induced chemical changes such as those described in Equations (1) and (2), must be established. However, a systematic determination of this one-on-one relationship is beyond the scope of this article. We defer this task to our future work and focus, in this article, on the evaluation of BDS as a potential platform to monitor the temporal changes during carbon capture.

Although typical desorption experiments take place at temperatures between 343 and 383 K,^{31,68,69} we chose $T = 340$ K in this study because it is low enough so we can observe all three relaxation processes within our frequency window to capture their changes upon CO_2 desorption while it is still higher than the potential melting temperature of PEI at $T = 325$ K and also allows for desorption to happen.⁶⁹

Figure 10A presents the dielectric spectra of a thin film of CO_2 -PEI exposed to a continuous pure N_2 atmosphere at selected times, $t = 0 - 6300$ s, and $T = 340$ K, where $t = 0$ s is the moment where the thin film CO_2 -PEI was exposed to pure N_2 (see inset of Figure 10A). Note that N_2 has little interaction with PEI. The dielectric measurements thus reflect the impact of CO_2 desorption on the dynamics of the thin PEI film. Significant changes can be observed in the overall dielectric dispersion that is primarily due to the shifts of the peak positions during CO_2 desorption. In particular, the process at around 10^4

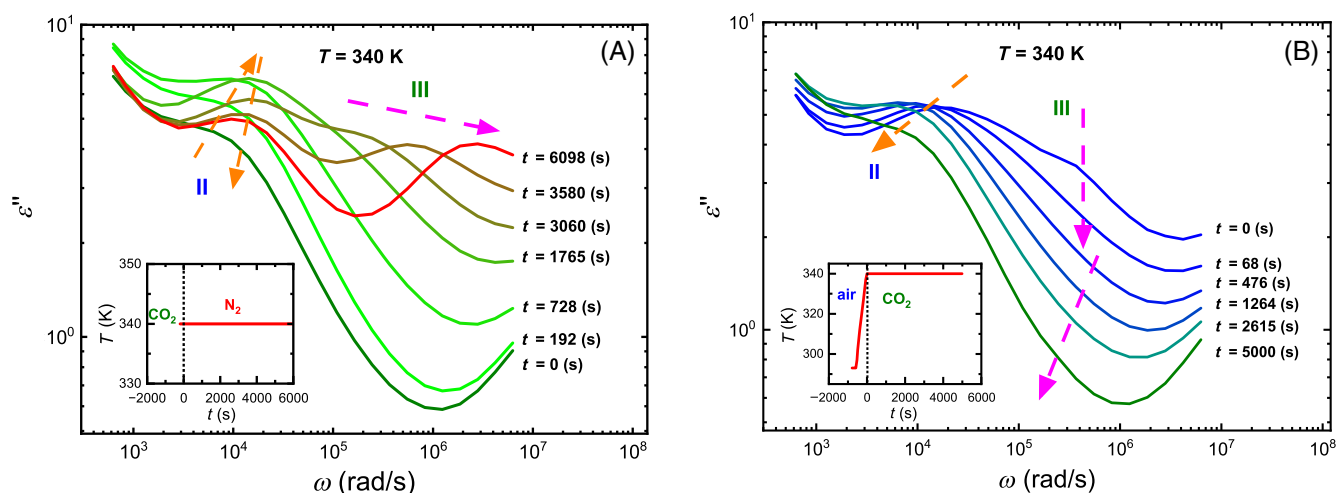


FIGURE 10 Dielectric loss spectra $\varepsilon''(\omega)$ of thin films of (A) CO₂-PEI that is exposed to pure N₂ and (B) air-PEI that is exposed to pure CO₂ at different times as indicated and recorded at 340 K. The insets depict the change of the surrounding gas as a function of time.

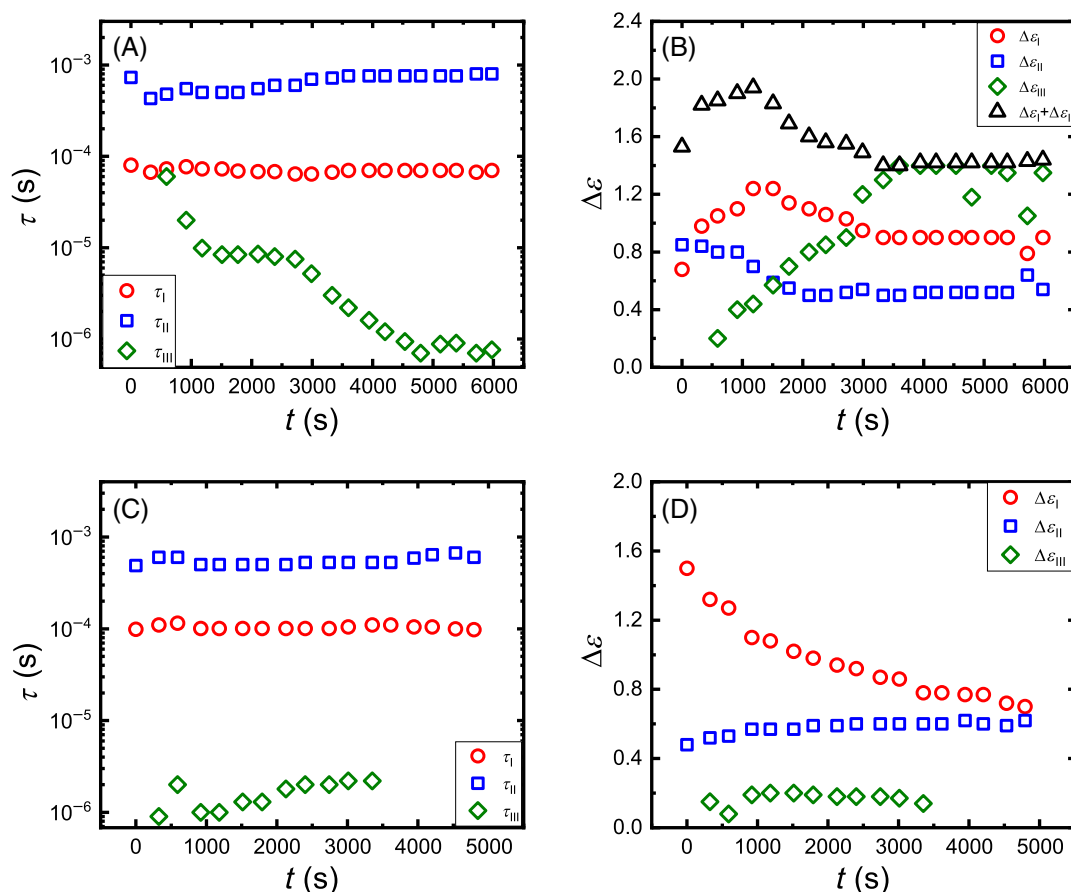


FIGURE 11 Temporal evolution of several relaxation parameters of thin PEI films during CO₂ adsorption and desorption: (A) mean relaxation time of thin film CO₂-PEI during CO₂ desorption, (B) the dielectric relaxation strengths of thin film CO₂-PEI during CO₂ desorption, (C) mean relaxation time of thin film air-PEI during CO₂ adsorption in pure CO₂, and (D) the dielectric relaxation strength of air-PEI during CO₂ adsorption in pure CO₂.

rad/s (process II) moves to a higher frequency and then slightly reduces upon CO₂ desorption. At the same time, the dielectric amplitude of this process first increases upon the initial stages of CO₂

desorption and decreases as the desorption continues, exhibiting an intriguing non-monotonic change in both the relaxation time and the dielectric relaxation strength (see the orange arrow in Figure 10A).

Remarkably, the process III becomes more and more visible upon CO_2 desorption, starting as a shoulder at the high-frequency wing of process II in the short-time limit and developing into a pronounced and well-resolved peak at higher frequencies in the long-time limit (the magenta arrow in Figure 10A). It is worth noting that the whole spectrum exhibits little further changes beyond $t = 5000$ s, implying the realization of a steady state in the pure N_2 atmosphere at this temperature.

Similarly, one can also use the BDS platform to characterize the kinetics of CO_2 sorption. Figure 10B depicts the influence of CO_2 on the dielectric properties of the thin film. Note in this particular case, the PEI film was initially exposed to a flow of ambient air (thin film air-PEI) where a dilute concentration of CO_2 is present. We switch from ambient air atmosphere to pure CO_2 at $t = 0$ s (see the inset of Figure 10B), representing the onset of additional CO_2 adsorption. The large changes in the dielectric spectra of the PEI film from exposure to dilute CO_2 to pure CO_2 , especially in process III, highlight the influence of the partial pressure of CO_2 on both the CO_2 sorption kinetics and the resulting PEI dynamics.

Opposite to the CO_2 desorption process, the relaxation processes move to lower frequencies with time along with a strong reduction in dielectric relaxation strength (the orange arrow of Figure 10B). In particular, the shoulder process at $\approx 3 \times 10^5$ rad/s (process III) at $t \approx 0$ moves toward the process at $\approx 1 \times 10^5$ rad/s (process II) and merges with its wing at $t > 4000$ s (the magenta arrow of Figure 10B). Little changes in the dielectric spectra beyond $t = 4000$ s suggest again the arrival of a steady state of the absorption process at long times. Therefore, time-resolved dielectric measurements can offer systematic characterizations of the kinetics CO_2 desorption and absorption, and thus are helpful to characterize the involved mechanisms and limitations in particular conditions like thin films or porous media.

Furthermore, an in-depth analysis of these dielectric measurements can provide quantitative characterizations of the kinetics of CO_2 desorption and adsorption. Figure 11A,C presents the temporal evolution of the relaxation times of the three processes, τ_I , τ_{II} , and τ_{III} , upon the CO_2 desorption and sorption processes, respectively; the

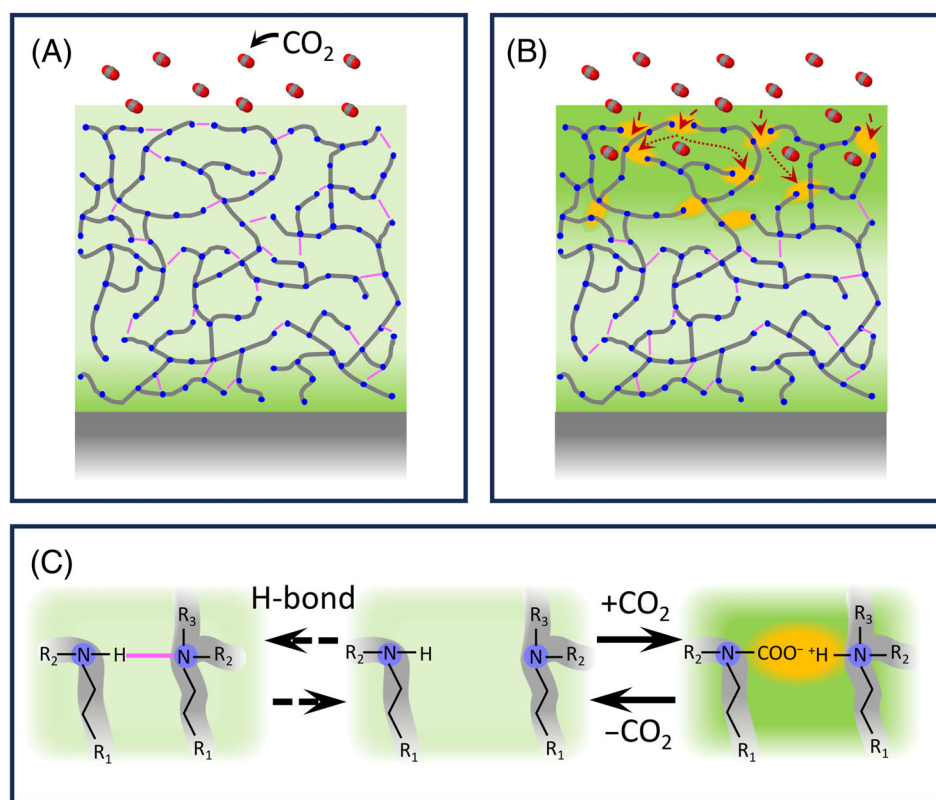


FIGURE 12 Schematic representations of a thin polyethylenimine (PEI) film on a substrate in different stages of CO_2 absorption: (A) completely desorbed film where solid grey lines are the PEI chains, blue dots are the amine groups along the chains, and the dashed magenta lines indicate the hydrogen (H)-bonding between amine groups; and (B) hypothesized partially CO_2 -absorbed film where the absorption of each CO_2 molecule physically cross-links two nearby amines through strong ionic interactions between ammonium and carbamate indicated by the yellow ellipses. These interactions strongly increase the segmental friction and slow down polymer dynamics (depicted as shading in darker green). In addition, the crosslinks impose steric constraints on the segmental motion both of which reduce the CO_2 diffusion into the film. Panel (C) shows a close-up of the transient H-bonding between adjacent amine groups in the desorbed state, as well as the much stronger physical cross-linking reaction through ionic interactions between ammonium and carbamate ions in the CO_2 -sorbed state (formed according to Equations 1 and 2); the same color code as in (A) and (B) is used, R_1 represents the continuing PEI chain while R_2 and R_3 are either PEI chains or terminal hydrogen.

corresponding development of the dielectric relaxation strengths, $\Delta\epsilon_I$, $\Delta\epsilon_{II}$, and $\Delta\epsilon_{III}$, is provided in Figure 11B,D.

In the CO₂ desorption/absorption process, the characteristic relaxation times of processes I and II change only slightly. However, their dielectric relaxation strengths $\Delta\epsilon$ have noticeable changes. In particular, $\Delta\epsilon_{II}$ reduces monotonically during CO₂ desorption while $\Delta\epsilon_I$ experiences an intriguing non-monotonic change. Note that although the absolute values of $\Delta\epsilon_I$ and $\Delta\epsilon_{II}$ bear quite a large uncertainty due to the strong superposition of both processes, the sum of them, $\Delta\epsilon_I + \Delta\epsilon_{II}$, is relatively robust and can be obtained directly from the changes of the storage permittivity at the corresponding frequencies. Specifically, the non-monotonic changes in the dielectric relaxation strengths indicate at least two competing mechanisms dictating the dielectric properties of PEI film at low frequencies during the CO₂ desorption/absorption.

The competing mechanisms could be due to a switch of the local molecular interactions between the amine groups and between the amine groups and CO₂ during CO₂ desorption or absorption. One possibility is the release of CO₂ that reduces the number of temporal “CO₂ cross-links” of the PEI film since one CO₂ molecule reacts with two primary or two secondary amine groups upon absorption completion (see Equations 1, 2, and Figure 12). The release of CO₂ thus alleviates the constraints imposed by the temporal cross-linking associated with CO₂ desorption, which yields an increment in the dielectric relaxation strength. Upon sufficient CO₂ desorption, the neighboring amine groups of PEI can re-establish new inter-segmental interaction, such as hydrogen bonding, that favors a reduction of the dielectric relaxation strength. Upon CO₂ sorption, the process is reversed where the progressing incorporation of CO₂ opens the existent hydrogen bonds among neighboring amine groups before the reaction between amines and CO₂ happens. As a consequence, $\Delta\epsilon_{II}$ or $\Delta\epsilon_I + \Delta\epsilon_{II}$ increases with time and then reduces. Future experiments and computer simulations might help to resolve the molecular origin of these interesting CO₂ desorption-induced dielectric property changes of PEI thin films (Figure 12).

More importantly, dramatic changes to process III upon CO₂ desorption or absorption occur. τ_{III} speeds up more than 100 times upon CO₂ desorption at $T = 340$ K. At the same time, $\Delta\epsilon_{III}$ increases from 0.2 to around 1.4 upon reaching a steady state. Conversely, absorption of CO₂ into the PEI film leads to a significant increment in τ_{III} and an obvious reduction in $\Delta\epsilon_{III}$. Note that $\Delta\epsilon_{III}$ does not exhibit a non-monotonic change with CO₂ desorption, implying a different physical origin of process III from process II. These changes in characteristic relaxation time and the dielectric relaxation strength agree well with the general expectation of the impact of CO₂ chemisorption to structural relaxation, where the interactions between PEI and CO₂ slow down the PEI dynamics and impose additional constraints on the dipolar relaxation of PEI (Figure 12). Furthermore, τ_{III} during CO₂ desorption and absorption saturates at around $t = 5000$ and 4000 s, respectively, indicating a long stabilization time required to reach the steady state of CO₂ desorption and a relatively slow CO₂ absorption rate at the testing condition. The temporal evolution of τ and $\Delta\epsilon$ during CO₂ absorption and desorption should contain key insights on the

CO₂ transport in PEI thin films. With such an experimental platform, future measurements in combination with simulations should be able to resolve intricate details such as the contributions of actual reaction rate and CO₂ transport through the film toward the amine binding sites.

4 | CONCLUSION

In summary, we have presented an investigation of the dynamics of PEI in bulk and thin films under the presence and in the absence of CO₂ using BDS and DSC. In both bulk and thin films, three dielectric relaxation processes have been observed in the studied temperature and frequency range, among them one is identified as the structural relaxation. The relaxations in the thin film are slower than their counterparts in bulk as expected due to the well-known interface effect and nanoconfinement effect. Furthermore, the CO₂-sorption leads to a noticeable shift in the T_g of the bulk PEI as demonstrated by both the BDS and DSC measurements. For thin film PEI, the slowing down of the structural relaxations due to CO₂ absorption can be more than 100 times at high temperatures of $T = 340$ K. We have attributed these large changes in dynamics to the CO₂-sorption-induced chemical changes to PEI.

To gain insights into the influence of amine-CO₂ reactions on PEI dynamics, we have conducted *in situ* BDS measurements of thin film PEI during CO₂ adsorption and desorption. Upon CO₂ desorption (or absorption), a progressively speeding up (or slowing down) in the high-frequency relaxation process (process III) along with a large increment (or reduction) in the dielectric amplitude has been observed, providing a detailed characterization of the impact of the CO₂ sorption kinetics on PEI dynamics. On the other hand, the relaxation process at the intermediate frequencies (process II) exhibits an interesting nonmonotonic shift in the relaxation time and the dielectric relaxation strength with CO₂ desorption, in contrast to the responses of the process III. Our measurements have also revealed large dynamics changes to thin film PEI while switching ongoing air (dilute CO₂ source) to pure CO₂, highlighting the effect of concentration of CO₂ (or the partial pressure of CO₂) on PEI dynamics. These results suggest an interesting interplay between the amine-CO₂ reactions and the dynamics of PEI. Furthermore, relaxation processes of different time scales can respond differently to the CO₂ absorption. Although the molecular origin of these temporal changes remains a topic of future investigation, these results highlight BDS can serve as a new platform for monitoring the CO₂ sorption-induced temporal dynamics changes of polyamine systems for carbon capture.

Furthermore, we would like to emphasize that, since the changes in the relaxation times and the dielectric relaxation amplitudes are due to the CO₂ desorption or absorption, one can in principle establish a correlation between the amount of CO₂ desorption or absorption with these dynamics changes. However, the current study focuses on demonstrating that BDS is a useful platform to track the evolution of PEI dynamics during carbon capture. With this in mind, we would also like to comment on how the revealed CO₂-sorption or CO₂-

desorption-induced PEI dynamics changes will help in both the understanding of the CO₂ capture performance of mesoporous substrate-supported polyamines and also guiding their future design for an improved functionality. As shown in Figure 11, the structural relaxation of PEI can slow down more than 100 times upon CO₂ sorption. Given the strong coupling between the CO₂ diffusivity and the structural relaxation of polymer membranes,⁷⁰ these observations might explain the often observed low amine efficiency of PEI films: the CO₂ sorption at the PEI/air surface slows down significantly the structural relaxation of the PEI network which in turn also slows down the CO₂ transport to the inner parts of the films.

Lastly, the dielectric measurements also point to the following two important factors: (i) a pronounced interface effect is expected due to the strong PEI/substrate interactions that largely elevate the glass transition temperature of the PEI film, demonstrated by the observation that the segmental dynamics of bulk PEI at $T = 283$ K are comparable to the thin film at $T = 340$ K (Figure 6B). Experiments have shown, that the polymer/substrate interactions play a major role in T_g -changes of thin polymer films.⁷¹⁻⁷³ In particular, introducing repulsive polymer/substrate interaction as well as decreasing the packing density at the interface can even reduce the T_g of the thin films.⁷³⁻⁷⁵ Thus, future designs of polyamine systems should also focus on the substrate with a target of reducing polyamine-substrate interactions. Our recent efforts to replace mesoporous silica-based substrates with porous polyolefin are along this line.³¹ (ii) The fact that the CO₂-chemisorption can connect neighboring amine groups through hydrogen-bonding and mimic the effect of chemical crosslinking temporarily should be considered in placing amine groups along the chains. When the chain lengths between neighboring crosslinks are small, these crosslinks can drastically hinder molecular motions, that is, slow down the dynamics (as schematically shown in Figure 12). From this perspective, one might also want to control the number density of the amine groups and the spatial distribution of the amines to alleviate the crosslinking effect on slowing down polymer dynamics which in turn would improve the CO₂ diffusivity.

AUTHOR CONTRIBUTIONS

Martin Tress: conceptualization; methodology; data curation; investigation; validation; formal analysis; funding acquisition; writing – review and editing; visualization. **Soma Ahmadi:** data curation; methodology; validation; formal analysis; writing – review and editing; visualization. **Shiwang Cheng:** conceptualization; methodology; data curation; investigation; validation; supervision; funding acquisition; project administration; resources; writing – original draft; writing – review and editing.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data used to create the displayed graphs is provided in the “Data.zip” file of Appendix S1. In particular, this contains dielectric permittivity, loss, and derivative spectra, $\epsilon'(\omega)$, $\epsilon''(\omega)$, and $-\frac{\pi}{2} \frac{\partial}{\partial \ln \omega} \epsilon'(\omega)$, respectively, at various temperatures of bulk v-PEI, bulk CO₂-PEI, bulk air-PEI (Figures 3, 4, 6, and Figure S2), thin film N₂-PEI (Figures 6, 7, 9, and Figure S1), thin film CO₂-PEI (Figures 8, 9, and Figure S1). Additionally, continuously recorded dielectric loss spectra at constant temperature (340 K) during desorption of thin film CO₂-PEI and absorption for thin film N₂-PEI are given (Figure 10). Further, the extracted characteristic relaxation times $\tau(T)$ and dielectric relaxation strengths $\Delta\epsilon(T)$ at various temperatures of bulk v-PEI, bulk CO₂-PEI, bulk air-PEI (Figure 4), and thin film N₂-PEI (Figure 7) are listed as well as these quantities as function of time $\tau(t)$ and $\Delta\epsilon(t)$ at constant temperature (340 K) during desorption of thin film CO₂-PEI and absorption for thin film N₂-PEI (Figure 11). The curves of the displayed fits to HN functions (Figures 4, 7, and 9) are also included. Furthermore, heat flow measurements (Figure 5) for bulk v-PEI, bulk CO₂-PEI, and bulk air-PEI are included. To verify the reliability of the data the reproducibility was confirmed by repeated measurements of several samples which showed similar results. Also, infrared spectra of CO₂-PEI are provided (Figure S3). The data that supports the findings of this article are available in the Appendix S1 of this article.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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